

Irrational Numbers and Infinite Decimal Expansions

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1 Introduction

A common intuitive way to think about irrational numbers is to imagine them as numbers whose decimal expansions continue indefinitely after the decimal point. For example,

$$\sqrt{2} = 1.4142135623730950488 \dots$$

and

$$\pi = 3.1415926535897932384 \dots$$

both have infinitely many decimal digits.

This intuition is useful, but it is not quite precise. Rational numbers can also have infinitely many digits after the decimal point. For instance,

$$\frac{1}{3} = 0.33333333 \dots$$

and

$$\frac{1}{7} = 0.142857142857142857 \dots$$

Thus, the mere fact that a decimal expansion is infinite does not distinguish irrational numbers from rational numbers.

The essential distinction is *periodicity*. Rational numbers have decimal expansions that eventually repeat, whereas irrational numbers have decimal expansions that never become periodic.

2 Decimal Expansions of Real Numbers

Every real number can be represented in decimal notation in the form

$$x = a_0.a_1a_2a_3 \dots,$$

where a_0 is an integer and each digit a_k belongs to the set

$$\{0, 1, 2, \dots, 9\}.$$

More formally, for a nonnegative real number we may write

$$x = a_0 + \sum_{n=1}^{\infty} \frac{a_n}{10^n}.$$

Thus the infinite string of decimal digits represents an infinite series. For example,

$$0.314159 \dots = \frac{3}{10} + \frac{1}{10^2} + \frac{4}{10^3} + \frac{1}{10^4} + \frac{5}{10^5} + \frac{9}{10^6} + \dots$$

Consequently, the decimal notation

$$a_0.a_1a_2a_3\dots$$

should not merely be regarded as a symbolic string. It is a compact notation for the limit of a sequence of finite decimal approximations.

For example,

$$\sqrt{2} = \lim_{n \rightarrow \infty} x_n,$$

where

$$x_1 = 1.4, \quad x_2 = 1.41, \quad x_3 = 1.414, \quad x_4 = 1.4142,$$

and so forth.

Each x_n is rational, but their limit may be irrational.

3 Terminating and Repeating Decimals

Some rational numbers have finite decimal representations. For example,

$$\frac{1}{2} = 0.5, \quad \frac{3}{8} = 0.375, \quad \frac{17}{20} = 0.85.$$

Such decimals are called *terminating decimals*.

However, even a terminating decimal may be written as an infinite decimal expansion by appending infinitely many zeros:

$$0.5 = 0.500000000\dots$$

There is also a second representation obtained by using infinitely many 9's:

$$0.5 = 0.499999999\dots$$

In particular,

$$1 = 0.999999999\dots$$

Thus decimal representations are not always unique. The only ambiguity occurs for terminating decimals: a terminating expansion ending in infinitely many zeros can also be represented by a corresponding expansion ending in infinitely many 9's.

Other rational numbers do not terminate but repeat periodically. For example,

$$\frac{1}{3} = 0.\overline{3},$$

where the bar means that the digit 3 repeats forever, and

$$\frac{1}{7} = 0.\overline{142857}.$$

The repeating block need not begin immediately after the decimal point. For example,

$$\frac{1}{6} = 0.1\overline{6}.$$

Here the digit 1 occurs once, after which the digit 6 repeats forever.

This motivates the notion of an *eventually periodic* decimal expansion.

A decimal expansion

$$a_0.a_1a_2a_3\dots$$

is eventually periodic if there exist positive integers N and p such that

$$a_{n+p} = a_n$$

for every $n \geq N$.

The number p is the length of a repeating block, while N marks the point after which the repetition begins.

4 The Characterization of Rational Numbers

One of the fundamental facts about positional notation is the theorem

$$x \in \mathbb{Q} \iff \text{the decimal expansion of } x \text{ is eventually periodic.}$$

Equivalently,

$$x \notin \mathbb{Q} \iff \text{the decimal expansion of } x \text{ is not eventually periodic.}$$

This theorem gives a precise version of the informal statement that irrational numbers have “non-repeating infinite decimals.”

The word *eventually* is important. A rational number may have an initial non-repeating segment. For example,

$$0.12345454545\dots = 0.123\overline{45}$$

is still eventually periodic and hence rational.

Conversely, an irrational decimal expansion may contain many repeated digits or repeated finite blocks. What is forbidden is not repetition itself, but the existence of a finite block that repeats forever from some point onward.

For example, an expansion of the form

$$0.1101001000100001000001\dots$$

contains many zeros and ones and exhibits a visible pattern, but it is not periodic. The number of zeros between successive ones keeps increasing. Therefore the expansion does not eventually repeat with any fixed period.

5 Why Rational Numbers Produce Repeating Decimals

The periodicity of rational decimal expansions can be understood directly from long division.

Consider a rational number

$$x = \frac{p}{q},$$

where $p, q \in \mathbb{Z}$ and $q > 0$.

When performing decimal long division, at each stage one obtains a remainder. The possible remainders are

$$0, 1, 2, \dots, q-1.$$

There are therefore only finitely many possible remainders.

If the remainder eventually becomes zero, the decimal expansion terminates.

Otherwise, the division process continues forever. Since there are only finitely many possible nonzero remainders, some remainder must eventually occur a second time. Once the same remainder occurs again, the subsequent long-division process repeats exactly as before. Hence the decimal digits from that point onward repeat periodically.

This is an application of the elementary principle that an infinite sequence taking values in a finite set must repeat some value.

Thus every rational number has an eventually periodic decimal expansion.

6 Why Repeating Decimals Are Rational

The converse direction is also straightforward.

Suppose, for example, that

$$x = 0.\overline{142857}.$$

Since the repeating block has six digits,

$$10^6 x = 142857.\overline{142857}.$$

Subtracting x gives

$$10^6 x - x = 142857,$$

and therefore

$$(10^6 - 1)x = 142857.$$

Hence

$$x = \frac{142857}{999999} = \frac{1}{7}.$$

The same argument works for every repeating decimal.

More generally, suppose

$$x = 0.\overline{d_1 d_2 \cdots d_p},$$

where $d_1 d_2 \cdots d_p$ denotes an integer D consisting of p digits. Then

$$10^p x = D + x,$$

so

$$x = \frac{D}{10^p - 1},$$

which is rational.

If the expansion contains some non-repeating digits before the periodic portion begins, multiplication by a suitable power of 10 first shifts those initial digits to the left of the decimal point. The same subtraction argument then applies.

Thus every eventually periodic decimal represents a rational number.

7 Irrational Numbers

An irrational number is, by definition, a real number that cannot be expressed as a quotient

$$\frac{p}{q}, \quad p, q \in \mathbb{Z}, \quad q \neq 0.$$

Since rational numbers are exactly those with eventually periodic decimal expansions, it follows that irrational numbers are exactly those whose decimal expansions never become periodic.

For example,

$$\sqrt{2} = 1.4142135623730950488016887 \dots$$

cannot eventually repeat, because $\sqrt{2}$ is irrational.

Similarly,

$$\pi = 3.1415926535897932384626433 \dots$$

and

$$e = 2.7182818284590452353602874 \dots$$

have non-eventually-periodic decimal expansions.

It is important to distinguish this statement from a much stronger claim such as “the digits of an irrational number are random.” Irrationality does not imply randomness.

One may deliberately construct an irrational number whose digits follow an entirely deterministic rule. For example, consider

$$x = 0.101001000100001000001 \dots,$$

where the number of zeros between successive ones increases by one each time.

The expansion clearly follows a simple deterministic construction. Nevertheless, it is not eventually periodic and therefore represents an irrational number.

Thus irrationality is fundamentally a statement about the impossibility of eventual periodicity, not about randomness or lack of structure.

8 Decimal Expansions Are Representations, Not the Numbers Themselves

There is another conceptual distinction worth making.

It is useful to visualize a real number by its infinite decimal expansion, but mathematically the number should not be identified too strongly with that particular digit string.

Decimal notation depends on our choice of base 10.

For example,

$$\frac{1}{3} = 0.333333 \dots_{10},$$

but in base 2,

$$\frac{1}{3} = 0.01010101 \dots_2.$$

The number itself is independent of either representation.

Likewise, the irrational number $\sqrt{2}$ has one sequence of digits in decimal notation, another in binary notation, another in hexadecimal notation, and so forth.

The distinction between rational and irrational numbers does not depend on the chosen base. For every integer base $b \geq 2$,

$$x \in \mathbb{Q} \iff \text{the base-}b \text{ expansion of } x \text{ is eventually periodic.}$$

Hence

$$x \notin \mathbb{Q} \iff \text{the base-}b \text{ expansion of } x \text{ is not eventually periodic.}$$

Decimal notation is therefore a representation of a deeper mathematical object: a real number.

9 Infinite Expansions and Limits

The idea of an infinite decimal expansion is also closely connected with the concept of a limit. Consider

$$x = 3.14159265\dots$$

Define the sequence of truncations

$$x_0 = 3,$$

$$x_1 = 3.1,$$

$$x_2 = 3.14,$$

$$x_3 = 3.141,$$

and so on.

Then

$$x = \lim_{n \rightarrow \infty} x_n.$$

Every finite truncation x_n is rational because it can be written as a fraction with denominator 10^n . Nevertheless, the limiting number may be irrational.

Thus an irrational number can be approached arbitrarily closely by rational numbers.

For example,

$$1.4, \quad 1.41, \quad 1.414, \quad 1.4142, \quad 1.41421, \quad \dots$$

are all rational, yet

$$\lim_{n \rightarrow \infty} x_n = \sqrt{2},$$

which is irrational.

This illustrates an important property of the real numbers: the rational numbers are dense in the real line. Between any two distinct real numbers there exists a rational number. Irrational numbers are also dense in the real line.

Consequently, the distinction between rational and irrational numbers is not a distinction between isolated regions of the real line. Rational and irrational numbers are interwoven arbitrarily closely.

10 Conclusion

It is reasonable to think of an irrational number as having an infinite string of digits after the decimal point, provided that the idea is stated carefully.

Merely having infinitely many decimal digits is not enough, because rational numbers such as

$$\frac{1}{3} = 0.333333 \dots$$

also have infinite decimal expansions.

The decisive property is eventual periodicity:

$$x \in \mathbb{Q} \iff x \text{ has an eventually periodic decimal expansion}$$

and therefore

$$x \notin \mathbb{Q} \iff x \text{ has a non-eventually-periodic decimal expansion.}$$

The infinite string of digits should also be understood as a representation of the number rather than as the number itself. Decimal notation is merely one positional numeral system among many.

A useful mental picture is therefore the following:

An irrational real number has an infinite positional expansion whose digits never settle into a finite repeating cycle.

This intuition is fully compatible with the formal definition of irrationality and also connects naturally with several central ideas of real analysis: infinite series, limits, completeness, and the density of rational and irrational numbers in the real line.